DOCUMENT 06

Development & Operations Documentation

Repository structure, coding standards, deployment runbook, CI/CD pipeline, observability, and disaster recovery.

SCOPE Repos • Standards • CI/CD • SLOs • DR • Backups

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Document 6 — Development & Operations Documentation

This document is the engineering how-to: README template, environment variables, coding standards, deployment runbook, CI/CD pipeline documentation, monitoring and alerting, and disaster recovery. It is written for engineers joining the project and for SREs operating the platform in production.

6.1 Repository Structure

The platform is a pnpm monorepo managed by Turborepo. Apps (portal, admin, marketing) live under /apps; NestJS services live under /services; shared packages (types, config, UI, domain primitives, test utils) live under /packages; infrastructure (Terraform, Helm, Argo CD manifests) lives under /infra; documentation (OpenAPI specs, ADRs, runbooks) lives under /docs. Each service is independently deployable and owns its own data.

6.2 Environment Variables (Selected)

VARIABLE	REQUIRED	DESCRIPTION	SOURCE
NODE_ENV	Yes	development | staging | production	Deployment env
DATABASE_URL	Yes	PostgreSQL connection string	Secrets Manager: aurumx/{env}/db/master
REDIS_URL	Yes	Redis cluster URL	Secrets Manager: aurumx/{env}/redis
KAFKA_BROKERS	Yes	Comma-separated Kafka broker list	Secrets Manager: aurumx/{env}/kafka
S3_DOCUMENTS_BUCKET	Yes	Bucket for KYC docs, contracts, assay certs	aurumx-{env}-documents-{region}
AWS_REGION	Yes	AWS region (af-west-1 prod / eu-west-1 DR)	Config
JWKS_URI	Yes	Auth0 JWKS endpoint	{JWT_ISSUER}.well-known/jwks.json

VARIABLE	REQUIRED	DESCRIPTION	SOURCE
DATADOG_API_KEY	Yes	Datadog ingestion key	Secrets Manager: aurumx/{env}/datadog
SENTRY_DSN	Yes	Sentry project DSN	Secrets Manager: aurumx/{env}/sentry
STANBIC_API_KEY	Yes	Stanbank escrow API key	Secrets Manager: aurumx/{env}/bank/stanbic
PMMC_API_KEY	Yes	PMMC export permit API	Secrets Manager: aurumx/{env}/pmmc
BOG_FX_API_KEY	Yes	Bank of Ghana FX approval API	Secrets Manager: aurumx/{env}/bog-fx
TWILIO_AUTH_TOKEN	Yes	Twilio WhatsApp + SMS	Secrets Manager: aurumx/{env}/twilio
AUDIT_LOG_HASH_CHAIN_SALT	Yes	Per-env salt for hash anchoring	Secrets Manager — read denied to SRE

6.3 Coding Standards

NAMING CONVENTIONS

ELEMENT	CONVENTION	EXAMPLE
Files (TypeScript)	kebab-case.ts	bank-adapter.ts
Folders	kebab-case	services/escrow/
Classes	PascalCase	TradeService, EscrowAggregate
Interfaces	PascalCase, no I prefix	EscrowProvider
Functions / methods	camelCase	fundEscrow(), validateKyc()
Variables	camelCase	tradeId, sellerOrgId
Constants	UPPER_SNAKE_CASE	MAX_NEGOTIATION_ROUNDS = 5
Database tables	snake_case, plural	organizations, gold_lots
Database columns	snake_case	seller_org_id, created_at
Kafka topics	\<domain\>.\<event-type\>.v\<n\>	trade.created.v1

ELEMENT	CONVENTION	EXAMPLE
API paths	kebab-case, plural nouns	/v1/gold-lots
JSON fields	camelCase	sellerOrgId, tradeId

GIT BRANCHING — TRUNK-BASED DEVELOPMENT

Trunk-based development with short-lived feature branches: branch from main with naming \<type\>/\<scope\>-\<short-desc\>; open PR against main; require 2 reviewers (Security & Compliance Engineer required for security-sensitive paths: services/escrow/, services/compliance/, services/audit-log/, infra/terraform/); CI must pass (lint, typecheck, unit, integration, e2e smoke, SAST, dependency scan, Terraform plan); squash-and-merge to main with Conventional Commit message; delete feature branch after merge.

CODE REVIEW CHECKLIST

- Functionality: does the change do what the PR says? Are edge cases covered?
- Tests: are unit + integration tests added? Do they exercise the new behavior?
- Architecture: respects service boundaries (Doc 4 §2.4)? No cross-service DB access?
- API contracts: if endpoints changed, OpenAPI spec updated? Breaking changes flagged?
- Data model: if schema changed, is there a backward-compatible migration?
- Idempotency: if endpoint has side effects, does it use Idempotency-Key?
- Security: no secrets? No SQL injection? RBAC enforced? Input validated?
- Observability: structured logs? Metrics emitted? Traces propagated?
- Performance: no N+1 queries? No unbounded result sets?
- Error handling: domain errors used? RFC 7807 format? No leaked stack traces?

6.4 Deployment Runbook

ENVIRONMENTS

ENVIRONMENT	PURPOSE	AWS ACCOUNT	REGION
dev (local)	Engineer local dev	N/A	N/A
staging	Pre-prod, UAT	aurumx-staging	af-west-1
prod	Production	aurumx-prod	af-west-1 (primary), eu-west-1 (DR)

DEPLOY TO PRODUCTION (MANUAL APPROVAL REQUIRED)

Production deploys require Change Advisory Board approval. Process: (1) Create release tag (e.g., v1.4.0); (2) GitHub Actions workflow builds container images, pushes to ECR, updates Helm values in infra/argocd/prod/, opens PR to argocd-config repo; (3) After PR approval + merge, Argo CD syncs to prod with rolling update (3-replica min, maxSurge=1, maxUnavailable=0); (4) Health checks (readiness probe) gate rollout; auto-rollback if rollout exceeds 5 min or readiness fails; (5) Post-deploy verification via Datadog APM error-rate check, Datadog Synthetics, and Playwright smoke tests against prod URL.

ROLLBACK PROCEDURE

Application rollback: Argo CD rollback to previous revision, or revert the PR that updated Helm values and let Argo CD auto-sync to reverted state. Database rollback is the last resort — forward-only migration model means we deploy a NEW migration that reverses the previous one; never restore from backup in prod unless data corruption is confirmed. Feature flag kill switch is the fastest rollback for behavior changes (LaunchDarkly toggle, effect global within 60 seconds).

6.5 CI/CD Pipeline

GitHub Actions + Argo CD pipeline. Stages: lint + format check → typecheck → unit tests + coverage gate → integration tests (Testcontainers) → SAST (Snyk) → dependency scan → container build + scan (Trivy) → e2e smoke (Playwright) → Terraform plan (if infra/ changed) → PR review (2+ reviewers) → merge to main → build + push to ECR → Argo CD auto-deploy to staging → smoke tests staging → manual release tag → CAB approval → Argo CD sync to prod → smoke tests prod → notify stakeholders.

QUALITY GATES

GATE	THRESHOLD	ENFORCEMENT
Unit test coverage (critical services)	≥ 90% lines	CI fails PR
Unit test coverage (other services)	≥ 80% lines	CI fails PR
E2E test coverage for trade flows	100% of UF-01 through UF-09	CI fails PR
Snyk SAST	No high or critical findings	CI fails PR (override only with Security Eng sign-off)
Trivy container scan	No critical vulnerabilities	CI fails deploy
Dependency scan	No high/critical CVEs unfixed \> 30 days	CI fails PR
Bundle size (PWA)	≤ 200 KB gzipped (initial route)	CI warns; blocks on regression \> 10%

GATE	THRESHOLD	ENFORCEMENT
Lighthouse PWA score	≥ 90 (perf, a11y, best practices, SEO)	CI warns on regression

6.6 Monitoring, Logging & Alerting

OBSERVABILITY STACK

PILLAR	TOOL	RETENTION	PURPOSE
Metrics	Datadog Metrics	15 months	Time-series metrics for infra + app
Logs	Datadog Logs	90 days hot, 2 years cold	Structured application + audit logs
Traces	Datadog APM (OpenTelemetry)	30 days	Distributed tracing across services
Errors	Sentry	90 days	Frontend + backend error aggregation
Uptime	Datadog Synthetics	30 days	Proactive endpoint checks
Real user monitoring	Datadog RUM	30 days	Frontend performance + errors
Security alerts	AWS GuardDuty + Datadog Cloud SIEM	13 months	Threat detection

KEY SLOS

SLO	TARGET	ERROR BUDGET (MONTHLY)
API availability (per region)	99.95%	22 min
API p95 latency (read endpoints)	≤ 500 ms	N/A (latency SLO)
API p95 latency (write endpoints, non-escrow)	≤ 1.5 s	N/A
API error rate (5xx)	\< 0.1%	N/A
Audit log integrity (hash chain)	100% (no breaks)	0 (zero tolerance)

SLO	TARGET	ERROR BUDGET (MONTHLY)
Kafka consumer lag (compliance)	\< 30 s	5 min/month
Escrow funding success rate	≥ 99.5%	22 min/month degraded
Compliance alert ack time (HIGH/CRITICAL)	≤ 1 hour, 95th percentile	5 events/month over SLA

ALERT TRIGGERS (SELECTED)

ALERT	TRIGGER CONDITION	SEVERITY
HighErrorRate	5xx error rate \> 1% for 5 min	SEV-2
PodCrashLooping	Any pod in CrashLoopBackOff for 5 min	SEV-2
DatabaseCPUHigh	RDS CPU \> 80% for 5 min	SEV-3
KafkaConsumerLagHigh	Lag \> 1000 messages for 5 min (compliance topic)	SEV-2
EscrowFundingFailure	Escrow funding failure rate \> 2% for 15 min	SEV-1
AuditLogHashChainBreak	ANY hash chain integrity check failure	SEV-1
Auth0Outage	Auth0 health check fails for 2 min	SEV-1
ComplianceAlertAckBreach	HIGH/CRITICAL compliance alert unack \> 1 hour	SEV-2
GuardDutyHighSeverityFinding	Any GuardDuty HIGH finding	SEV-1

6.7 Disaster Recovery & Backup

BACKUP STRATEGY

DATA TYPE	BACKUP METHOD	FREQUENCY	RETENTION
PostgreSQL (RDS)	Automated snapshot + PITR (Point-in-Time Recovery)	Daily snapshots; PITR every 5 min	Snapshots 30 days; PITR 14 days

DATA TYPE	BACKUP METHOD	FREQUENCY	RETENTION
PostgreSQL (cross-region)	Cross-region automated snapshot copy	Daily	30 days in eu-west-1
S3 documents (KYC, contracts, assay certs)	S3 Cross-Region Replication (CRR)	Continuous	7 years (regulatory)
S3 audit log (WORM)	S3 Object Lock (Compliance mode)	On write	7 years
Kafka audit log topic	Log compaction + S3 archival	Continuous	7 years
EKS cluster state	Argo CD GitOps (state in Git) + EKS etcd snapshot	Continuous (Git) + daily (etcd)	Git: indefinite; etcd: 30 days

RTO AND RPO

TIER	WORKLOAD	RTO	RPO
Tier 0	Trade, Escrow, Audit Log	≤ 4 hours	≤ 5 minutes
Tier 1	Member, RFQ, Auction, Negotiation	≤ 8 hours	≤ 15 minutes
Tier 2	Analytics, Notification, Export Doc	≤ 24 hours	≤ 1 hour
Tier 3	Marketing site, internal tools	≤ 48 hours	≤ 24 hours

CROSS-REGION DR FAILOVER (FULL REGION LOSS)

Triggers: AWS region outage; regional disaster; major security incident requiring full region rebuild. Procedure: (1) Declare DR activation (Vendor PM + Chamber Exec); (2) Promote DR read replica in eu-west-1; (3) Restore S3 documents (CRR is continuous; verify count); (4) Start Kafka MirrorMaker2 to bridge events from primary region; (5) Update DNS (Route 53) to point api.aurumx.gh to DR region's ALB; (6) Verify DR endpoints; (7) Notify all stakeholders. Estimated RTO: 2–4 hours (most of which is DNS propagation + verification).